CNC PROGRAM SHEET

Part: Botlabs Hub | Rev: A | Date: 2026-06-24

Programmer: ShiaanX-AI

Material: ALUMINIUM

Max RPM: 10000 Coolant: through spindle

Setups: 4

COORDINATE SYSTEM & WORK ZERO (G54)

Work zero convention	--	**Part height (Y)**	-- mm
CAD up-axis	--	**X range (machine)**	-- to -- mm
Work zero in CAD	X=0 Y=0 Z=0	**Z range (machine)**	-- to -- mm
G54 meaning	Set controller G54 offset so that X0 Y0 Z0 = top face at origin	**Safe rapid Y**	10 mm above work zero

TOOL LIST

T#	Tool ID	Description	Dia (mm)	Manufacturer	Grade	Used for
T01	EM_2FL_1MM	2-flute solid carbide end mill 1mm â€” aluminium geometry	1.0	Sandvik Coromant	1630	contour_mill
T02	DRILL_1.2MM	Solid carbide jobber drill 1.2mm	1.2	Sandvik Coromant	H10F	twist_drill
T03	DRILL_1.25MM	Solid carbide jobber drill 1.25mm â€” nearest standard to 1.26mm	1.25	Sandvik Coromant	H10F	twist_drill
T04	DRILL_1.6MM	Solid carbide jobber drill 1.6mm	1.6	Sandvik Coromant	H10F	twist_drill
T05	DRILL_1.8MM	Solid carbide jobber drill 1.8mm	1.8	Sandvik Coromant	H10F	twist_drill
T06	EM_2FL_2MM	2-flute solid carbide end mill 2mm â€” aluminium geometry	2.0	Sandvik Coromant	1630	counterbore_mill
T07	DRILL_2.0MM	Solid carbide jobber drill 2.0mm	2.0	Sandvik Coromant	H10F	twist_drill
T08	SPOT_090_3MM	90-degree solid carbide spot drill, 3mm shank	3.0	Sandvik Coromant	H10F	spot_drill
T09	EM_2FL_3MM	2-flute solid carbide end mill 3mm â€” aluminium geometry	3.0	Sandvik Coromant	1630	counterbore_mill
T10	DRILL_3.0MM	Solid carbide jobber drill 3.0mm	3.0	Sandvik Coromant	H10F	twist_drill
T11	SPOT_090_4MM	90-degree solid carbide spot drill, 4mm shank	4.0	Sandvik Coromant	H10F	spot_drill
T12	EM_2FL_4MM	2-flute solid carbide end mill 4mm â€” aluminium geometry	4.0	Sandvik Coromant	1630	contour_mill
T13	EM_2FL_20MM	2-flute solid carbide end mill 20mm â€” aluminium geometry	20.0	Sandvik Coromant	1630	circular_interp
T14	FACEMILL_50MM	50mm indexable face mill / shell mill â€” 4 inserts	50.0	Sandvik Coromant		face_mill

SETUP 1 -- -Y approach (bottom — part flipped upside down)

PRINCIPAL | 16 clusters | 29 operations

Fixture	Step-jaw vise or fixture plate with bosses as datums.
Description	Flipped setup — part inverted so bottom face is accessible to spindle. Clamp on previously machined top features or use step-jaw vise.
Rotation	None -- standard orientation
WCS	G54
Origin X/Y/Z	X=-- Y=-- Z=--
Face (W x H)	--
Max depth	--

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU CENTER DRILL	T08	3.0	0	--	10000	500	--	--	--	00:00
2	ALU 1.25 DRILL	T03	1.25	0	1.0	10000	180	1.0	--	--	00:00
3	ALU CENTER DRILL	T08	3.0	0	--	10000	500	--	--	--	00:00
4	ALU 1.25 DRILL	T03	1.25	0	1.0	10000	180	1.0	--	--	00:00
5	ALU CENTER DRILL	T11	4.0	0	--	10000	600	--	--	--	00:00
6	ALU 3 DRILL	T10	3.0	0	0.5	10000	450	0.5	--	--	00:00
7	ALU CENTER DRILL	T11	4.0	0	--	10000	600	--	--	--	00:00
8	ALU 3 DRILL	T10	3.0	0	0.5	10000	450	0.5	--	--	00:00
9	ALU CENTER DRILL	T08	3.0	0	--	10000	500	--	--	--	00:00
10	ALU 2 DRILL	T07	2.0	0	1.95	10000	320	1.95	--	--	00:00
11	ALU CENTER DRILL	T08	3.0	0	--	10000	500	--	--	--	00:00
12	ALU 2 DRILL	T07	2.0	0	1.95	10000	320	1.95	--	--	00:00
13	ALU CENTER DRILL	T08	3.0	0	--	10000	500	--	--	--	00:00
14	ALU 2 DRILL	T07	2.0	0	1.95	10000	320	1.95	--	--	00:00
15	ALU CENTER DRILL	T08	3.0	0	--	10000	500	--	--	--	00:00
16	ALU 2 DRILL	T07	2.0	0	1.95	10000	320	1.95	--	--	00:00
17	ALU 50 FACEMILL BORE	T14	50.0	0	0.66	5090	1527	0.661	--	--	00:00
18	ALU 20 ENDMILL BORE	T13	20.0	0	0.66	7960	955	0.661	--	--	00:00
19	ALU CENTER DRILL	T11	4.0	0	--	10000	600	--	--	--	00:00
20	ALU 3 DRILL	T10	3.0	0	0.5	10000	450	0.5	--	--	00:00
21	ALU CENTER DRILL	T11	4.0	0	--	10000	600	--	--	--	00:00
22	ALU 3 DRILL	T10	3.0	0	0.5	10000	450	0.5	--	--	00:00
23	ALU CENTER DRILL	T08	3.0	0	--	10000	500	--	--	--	00:00
24	ALU 2 DRILL	T07	2.0	0	0.29	10000	320	0.295	--	--	00:00
25	ALU CENTER DRILL	T08	3.0	0	--	10000	500	--	--	--	00:00
26	ALU 2 DRILL	T07	2.0	0	0.29	10000	320	0.295	--	--	00:00
27	ALU 4 ENDMILL BOSS	T12	4.0	0	0.41	10000	360	0.414	--	--	00:00

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
28	ALU 4 ENDMILL BOSS	T12	4.0	0	0.41	10000	360	0.414	--	--	00:00
29	ALU 50 FACEMILL FACE	T14	50.0	0	--	5090	1527	2.0	--	--	00:00

ESTIMATED SETUP TIME: 00:00 (includes tool changes)

SETUP 2 -- +Y approach (top — default VMC position)

PRINCIPAL | 9 clusters | 18 operations

Fixture	Standard vise or fixture plate, jaws on side faces.
Description	Standard top-face setup. Clamp part with top face accessible to spindle. Machine all features in this setup before repositioning.
Rotation	None -- standard orientation
WCS	G54
Origin X/Y/Z	X=-- Y=-- Z=--
Face (W x H)	--
Max depth	--

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU CENTER DRILL	T08	3.0	0	--	10000	500	--	--	--	00:00
2	ALU 1.8 DRILL	T05	1.8	0	1.86	10000	280	1.856	--	--	00:00
3	ALU CENTER DRILL COUNTERBORE	T08	3.0	0	--	10000	500	--	--	--	00:00
4	ALU 1.25 DRILL COUNTERBORE	T03	1.25	0	8.0	10000	180	8.0	--	--	00:00
5	ALU 3 ENDMILL COUNTERBORE	T09	3.0	0	--	10000	300	3.0	--	--	00:00
6	ALU 3 ENDMILL COUNTERBORE	T09	3.0	0	--	10000	300	3.0	--	--	00:00
7	ALU CENTER DRILL COUNTERBORE	T08	3.0	0	--	10000	500	--	--	--	00:00
8	ALU 1.25 DRILL COUNTERBORE	T03	1.25	0	8.0	10000	180	8.0	--	--	00:00
9	ALU 3 ENDMILL COUNTERBORE	T09	3.0	0	--	10000	300	3.0	--	--	00:00
10	ALU 3 ENDMILL COUNTERBORE	T09	3.0	0	--	10000	300	3.0	--	--	00:00
11	ALU CENTER DRILL	T08	3.0	0	--	10000	500	--	--	--	00:00
12	ALU 1.6 DRILL	T04	1.6	0	1.0	10000	240	1.0	--	--	00:00
13	ALU CENTER DRILL	T08	3.0	0	--	10000	500	--	--	--	00:00
14	ALU 1.2 DRILL	T02	1.2	0	2.04	10000	180	2.044	--	--	00:00
15	ALU 1 ENDMILL BOSS	T01	1.0	0	0.6	10000	100	0.6	--	--	00:00
16	ALU 1 ENDMILL BOSS	T01	1.0	0	0.6	10000	100	0.6	--	--	00:00
17	ALU 1 ENDMILL BOSS	T01	1.0	0	0.6	10000	100	0.6	--	--	00:00
18	ALU 1 ENDMILL BOSS	T01	1.0	0	0.6	10000	100	0.6	--	--	00:00

ESTIMATED SETUP TIME: 00:00 (includes tool changes)

SETUP 3 -- angled approach [0.68, -0.0, 0.733]

ANGLED | 1 clusters | 4 operations

Fixture	Sine plate or angled fixture set to 90.0°, or 4th/5th axis rotary table if available.
Description	Angled setup. Tilt part 90.0° around compound axis [-0.733, 0.0, 0.68] — use sine plate or 5-axis fixture from the default orientation. Veri
Rotation required	90.0 deg around compound axis [-0.733, 0.0, 0.68] — use sine plate or 5-axis fixture -- verify with dial indicator before cutting
WCS	G54
Origin X/Y/Z	X=-- Y=-- Z=--
Face (W x H)	--
Max depth	--

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	**FIXTURE ROTATION -- Feature axis [-0.68, 0.0, -0.733] is not aligned to a principal axis — part must be rotated or an angled fixture used before drilling**										
2	ALU CENTER DRILL COUNTERBORE ANGLED	T08	3.0	0	--	10000	500	--	--	--	00:00
3	ALU 1.25 DRILL COUNTERBORE ANGLED	T01	1.25	0	2.1	10000	180	2.1	--	--	00:00
4	ALU 2 ENDMILL COUNTERBORE ANGLED	T06	2.0	0	--	10000	200	2.0	--	--	00:00

ESTIMATED SETUP TIME: 00:00 (includes tool changes)

SETUP 4 -- angled approach [-0.68, -0.0, 0.733]

ANGLED | 1 clusters | 4 operations

Fixture	Sine plate or angled fixture set to 90.0°, or 4th/5th axis rotary table if available.
Description	Angled setup. Tilt part 90.0° around compound axis [-0.733, -0.0, -0.68] — use sine plate or 5-axis fixture from the default orientation. Ve
Rotation required	90.0 deg around compound axis [-0.733, -0.0, -0.68] — use sine plate or 5-axis fixture -- verify with dial indicator before cutting
WCS	G54
Origin X/Y/Z	X=-- Y=-- Z=--
Face (W x H)	--
Max depth	--

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	**FIXTURE ROTATION -- Feature axis [-0.68, 0.0, -0.733] is not aligned to a principal axis — part must be rotated or an angled fixture used before drilling**										
2	ALU CENTER DRILL COUNTERBORE ANGLED	T08	3.0	0	--	10000	500	--	--	--	00:00
3	ALU 1.25 DRILL COUNTERBORE ANGLED	T01	1.25	0	2.1	10000	180	2.1	--	--	00:00
4	ALU 2 ENDMILL COUNTERBORE ANGLED	T06	2.0	0	--	10000	200	2.0	--	--	00:00

ESTIMATED SETUP TIME: 00:00 (includes tool changes)

WARNINGS & FLAGGED ITEMS

Type	Cluster	Operation	Detail
RPM CAPPED	C0	spot_drill	Effective Vc = 94.2 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C0	twist_drill	Effective Vc = 56.5 m/min (reduced from 110 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C1	spot_drill	Effective Vc = 94.2 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C1	twist_drill	SUBSTITUTION: 1.26mm is non-standard. Using nearest standard size 1.25mm. Verify hole tolerance before confirming.
RPM CAPPED	C1	twist_drill	Effective Vc = 39.3 m/min (reduced from 90 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C1	counterbore_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C1	counterbore_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C2	spot_drill	Effective Vc = 94.2 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C2	twist_drill	SUBSTITUTION: 1.26mm is non-standard. Using nearest standard size 1.25mm. Verify hole tolerance before confirming.
RPM CAPPED	C2	twist_drill	Effective Vc = 39.3 m/min (reduced from 90 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C2	counterbore_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C2	counterbore_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C3	spot_drill	Effective Vc = 94.2 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C3	twist_drill	SUBSTITUTION: 1.26mm is non-standard. Using nearest standard size 1.25mm. Verify hole tolerance before confirming.
RPM CAPPED	C3	twist_drill	Effective Vc = 39.3 m/min (reduced from 90 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C4	spot_drill	Effective Vc = 94.2 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C4	twist_drill	SUBSTITUTION: 1.26mm is non-standard. Using nearest standard size 1.25mm. Verify hole tolerance before confirming.
RPM CAPPED	C4	twist_drill	Effective Vc = 39.3 m/min (reduced from 90 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C5	spot_drill	Effective Vc = 125.7 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C5	twist_drill	Effective Vc = 94.2 m/min (reduced from 140 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C6	spot_drill	Effective Vc = 125.7 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C6	twist_drill	Effective Vc = 94.2 m/min (reduced from 140 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C7	spot_drill	Effective Vc = 94.2 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C7	twist_drill	SUBSTITUTION: 1.26mm is non-standard. Using nearest standard size 1.25mm. Verify hole tolerance before confirming.
RPM CAPPED	C7	twist_drill	Effective Vc = 39.3 m/min (reduced from 90 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C7	counterbore_mill	Effective Vc = 62.8 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C8	spot_drill	Effective Vc = 94.2 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C8	twist_drill	SUBSTITUTION: 1.26mm is non-standard. Using nearest standard size 1.25mm. Verify hole tolerance before confirming.
RPM CAPPED	C8	twist_drill	Effective Vc = 39.3 m/min (reduced from 90 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C8	counterbore_mill	Effective Vc = 62.8 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C9	spot_drill	Effective Vc = 94.2 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C9	twist_drill	Effective Vc = 50.3 m/min (reduced from 100 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C10	spot_drill	Effective Vc = 94.2 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C10	twist_drill	Effective Vc = 37.7 m/min (reduced from 90 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C11	spot_drill	Effective Vc = 94.2 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C11	twist_drill	Effective Vc = 62.8 m/min (reduced from 120 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C12	spot_drill	Effective Vc = 94.2 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C12	twist_drill	Effective Vc = 62.8 m/min (reduced from 120 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C13	spot_drill	Effective Vc = 94.2 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.

Type	Cluster	Operation	Detail
RPM CAPPED	C13	twist_drill	Effective Vc = 62.8 m/min (reduced from 120 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C14	spot_drill	Effective Vc = 94.2 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C14	twist_drill	Effective Vc = 62.8 m/min (reduced from 120 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C16	spot_drill	Effective Vc = 125.7 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C16	twist_drill	Effective Vc = 94.2 m/min (reduced from 140 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C17	spot_drill	Effective Vc = 125.7 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C17	twist_drill	Effective Vc = 94.2 m/min (reduced from 140 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C18	spot_drill	Effective Vc = 94.2 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C18	twist_drill	Effective Vc = 62.8 m/min (reduced from 120 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C19	spot_drill	Effective Vc = 94.2 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C19	twist_drill	Effective Vc = 62.8 m/min (reduced from 120 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C20	contour_mill	Effective Vc = 31.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C21	contour_mill	Effective Vc = 31.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C22	contour_mill	Effective Vc = 31.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C23	contour_mill	Effective Vc = 31.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C24	contour_mill	Effective Vc = 125.7 m/min (reduced from 380 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C25	contour_mill	Effective Vc = 125.7 m/min (reduced from 380 m/min). Feed rate adjusted accordingly.

SETUP 1 — FIXTURE SPECIFICATION WCS: G54 | -Y approach (bottom â€” part flipped upside down)

Field	Value
Fixture Type	Vise (Standard Hard Jaws)
Jaw Type	Standard Hard Jaws
Jaw Opening	45 mm
Jaw Height	38 mm
Parallels	None (part sits on jaw serrations)
Clamped Faces	+X, -X
Rest Face	-Z
Datum Face	+Y
Z Zero Reference	Top Of Stock
WCS XY Zero	Center Of Part
Clamp Clearance	2.0 mm
Soft Jaws Required	No

SETUP NOTES

- Zero off top face of raw billet
- Jaw opening set to 45mm â€” verify engagement before cutting

SETUP 2 — FIXTURE SPECIFICATION WCS: G54 | +Y approach (top â€” default VMC position)

Field	Value
Fixture Type	Vise (Standard Hard Jaws)
Jaw Type	Standard Hard Jaws
Jaw Opening	45 mm
Jaw Height	38 mm
Parallels	None (part sits on jaw serrations)
Clamped Faces	+X, -X
Rest Face	-Z
Datum Face	-Y
Z Zero Reference	Top Of Stock
WCS XY Zero	Center Of Part
Clamp Clearance	2.0 mm
Soft Jaws Required	No

SETUP NOTES

- Zero off top face of raw billet
- Jaw opening set to 45mm â€” verify engagement before cutting

SETUP 3 — FIXTURE SPECIFICATION WCS: G54 | angled approach [0.68, -0.0, 0.733]

Field	Value
Fixture Type	Angle Plate
Jaw Type	Standard Hard Jaws
Jaw Opening	45 mm
Jaw Height	38 mm
Parallels	None (part sits on jaw serrations)
Clamped Faces	+X, -X
Rest Face	-Z
Datum Face	-Z
Z Zero Reference	Top Of Stock
WCS XY Zero	Center Of Part
Clamp Clearance	2.0 mm
Soft Jaws Required	No

SETUP NOTES

- Zero off top face of raw billet
- Jaw opening set to 45mm â€” verify engagement before cutting
- Angle plate setup â€” verify squareness to spindle before zeroing

SETUP 4 — FIXTURE SPECIFICATION WCS: G54 | angled approach [-0.68, -0.0, 0.733]

Field	Value
Fixture Type	Angle Plate
Jaw Type	Standard Hard Jaws
Jaw Opening	45 mm
Jaw Height	38 mm
Parallels	None (part sits on jaw serrations)
Clamped Faces	+X, -X
Rest Face	-Z
Datum Face	-Z
Z Zero Reference	Top Of Stock
WCS XY Zero	Center Of Part
Clamp Clearance	2.0 mm
Soft Jaws Required	No

SETUP NOTES

- Zero off top face of raw billet
- Jaw opening set to 45mm â€" verify engagement before cutting
- Angle plate setup â€" verify squareness to spindle before zeroing